Ram Physical Coordinates and Stable Extensions for Critical FK–Ising Walk Limits

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Abstract

We examine the pinned manuscript on stationary random walks on critical FK–Ising maps and Liouville Brownian motion. The exact state space, corner mass, all-edge energy, centered inverse, and deterministic attempt clock are retained in a physical Ram representation. We reconstruct the source's route from its explicitly imported common-coordinate and unit-conductivity hypotheses through positive-time operators, cutoff recovery, stationary tightness, and the cable-path limit. We prove a graph- and positive-parameter-uniform resolvent perturbation bound, and a conditional weighted path-limit extension for uniformly vanishing conductance perturbations with the same n-clock. We also prove a conditional extension to initial densities converging in the common corner-time L^1 space, including unbounded densities, give constructive corner-average realizations, and quantify transfer of path events by uniform integrability. The extension uses classical change of initial law, not a new quantum-geometric theorem. Exact multigraph and density tests check finite identities without purporting to verify the FK scaling limit. No arbitrary-vertex result, global proof replacement, historical novelty, or computational speedup is claimed. Contribution assessment: C2/5, the quantitative finite perturbation bound and conditional weighted and initial-density path extensions add limited improvements without replacing the imported geometric and electrical inputs.

1 Scope, Evidence, and the Precise Question

The target is OpenAI's October 5, 2026 manuscript [1], frozen at the following commit:

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

We read its actual `main.tex`, all five mathematical sections (introduction, inputs, operators, tightness, paths), and bibliography. Statements imported from the spectral companion [2] remain hypotheses for the present report. Selected actual companion passages were read to check conventions and domains, not to certify that companion's entire proof or its foundational dependencies.

There are three logically different layers. The *source assertion* is a joint metric-measure and time-parametrized walk convergence theorem. The *local reconstruction* below explains its dependence on a specific collection of strong electrical and geometric inputs. The *present contribution* is a finite perturbation bound, conditional weighted and initial-law extensions, and its exact representation and testing account. Neither algebraic repackaging nor finite successful checks establish the source assertion.

The private material was used at actual mathematical passages: Volume 21's physical-metric chapter [3] supplies weighted Gram matrices, physical symmetry, and omitted residuals; its spectral-limit chapter [4] separates strong from norm resolvents and insists on operator domains. The gauge and elimination passages of [5] explain the distinction between zero-sum loads, mass-centered potentials, and transported dual loads. These are representation principles, not an independently proved FK universality theorem. Classical Hilbert-space and Markov-process mathematics is credited as such throughout.

Two assessment scales. The initial catalogue records relevance $L4/5$: the corpus directly addresses the physical form/resolvent objects at issue. Contribution $C2/5$ measures realized improvement relative to this pinned source, not relevance, report length, or the importance of the source theorem. Here $C1$ means none demonstrated, $C2$ a limited contribution, $C3$ a notable one, $C4$ a substantial one, and $C5$ a very high contribution. The claims in Sections 5–7 are bounded conditional additions; the source's imported results earn no new credit.

2 The Source Problem Space

2.1 Random maps, walks, and the mass measure

For every integer $n \geq 1$, let $\mathcal{F}_n$ be the finite set of rooted pairs (M, A), with M a connected n-edge planar multigraph embedded in the oriented sphere, root an oriented edge, and $A \subset E(M)$. Equivalence preserves the orientation, root, and decoration. Write $A^* = \{e^* : e \notin A\}$ in the dual map, and let k count connected components including isolated vertices. The law is

$$\mathbb{P}((M_n, A_n) = (M, A)) = Z_n^{-1} 2^{[k(A)+k(A^*)-1]/2}. \tag{2.1}$$

This is an annealed law on maps and decorations. Once the map is fixed, the walk uses *all original edges*, not only A_n. At a vertex it chooses uniformly from incident half-edges; loops have two such half-edges and parallel edges are distinct choices. The mass is

$$\mu_n(v) = \frac{d_v}{2n}, \qquad d_v = \deg(v), \qquad H_n = L^2(V_n, \mu_n), \qquad M_n^{\mathrm{mass}} = \mathrm{diag}(\mu_n(v)). \tag{2.2}$$

The map symbol M_n and the explicitly superscripted mass matrix should not be confused. Connectedness and $n \geq 1$ make all displayed masses positive, including the one-vertex graph with loops only.

Let m_{xy} count nonloop edges for $x \neq y$, and let ℓ_x count loops at x. Then

$$P_n(x, y) = \frac{m_{xy}}{d_x} \ (x \neq y), \qquad P_n(x, x) = \frac{2\ell_x}{d_x}, \qquad G_n = n(I - P_n).$$

The generator of the continuous-time chain is $-G_n$, with total *attempt* rate n, not necessarily total state-changing rate n. All functions on V_n lie in both the form and operator domains. Detailed balance is $\mu_n(x)P_n(x, y) = \mu_n(y)P_n(y, x)$.

Choose one orientation for each original edge and let D_n be its edge-by-vertex incidence matrix; a loop has a zero row. The raw electrical matrix is $C_n = D_n^T D_n$. Direct multiplication yields

$$M_n^{\mathrm{mass}} G_n = \tfrac{1}{2} C_n, \qquad \mathcal{E}_n(u, v) = \langle u, G_n v \rangle_{\mu_n} = \tfrac{1}{2} \sum_{e \in E(M_n)} d_e u \, d_e v. \tag{2.3}$$

Indeed each oriented half-edge has stationary accelerated weight $[d_x/(2n)](n/d_x) = 1/2$. Pair the two ends of a nonloop edge; the result is half its difference product. Both ends of a loop give zero. The kernel is exactly the constants because a zero-energy function is constant on every edge of a connected graph. This is the source's graph-energy lemma, not a new result of this report.

2.2 Continuum space and operator domains

The limiting object is the *ordinary* area-one $\sqrt{3}$-quantum sphere, obtained by disintegrating the sphere law at total area one and forgetting the distinguished points. Its area probability is μ_h, its intrinsic metric is D_h in a fixed deterministic normalization, and its conformal coordinate space is S^2 with round metric ρ. These are different metric objects: D_h is not silently identified with the intrinsic pseudometric of the Dirichlet form.

Let $\mathcal{E}^0(f,g) = \int_{S^2} \nabla f \cdot \nabla g \, \mathrm{dvol}_0$. The companion's Proposition 8.3 states, with its quasi-continuous trace R_{μ_h}, that the closed half-energy form has

$$\mathcal{D}_h = R_{\mu_h}(H^1(S^2, \mathrm{vol}_0)), \qquad \mathcal{E}_h(R_{\mu_h} f, R_{\mu_h} g) = \tfrac{1}{2}\mathcal{E}^0(f,g). \tag{2.4}$$

Its nonnegative self-adjoint generator G on $H_h = L^2(\mu_h)$ has the weak domain

$$D(G) = \{u \in \mathcal{D}_h : \exists b \in H_h \ \forall v \in \mathcal{D}_h, \mathcal{E}_h(u,v) = \langle b, v\rangle_{\mu_h}\}, \qquad Gu = b. \tag{2.5}$$

The real Hilbert convention is used; complex functions can be treated with conjugation. This is neither a classical pointwise formula $-\frac{1}{2}e^{-\sqrt{3}h}\Delta$ nor an assertion that every smooth coordinate function belongs to $D(G)$. Smooth cutoffs belong to the *form* domain. Their resolvent smoothings belong to $D(G)$. The closed form is regular, and the associated diffusion is Liouville Brownian motion (LBM) in the half-energy clock. These facts, including the trace identification, are imported.

2.3 Exact quantifiers of the headline assertion

Let $a_n > 0$ be any deterministic sequence tending to zero such that

$$(V_n, a_n d_n, \mu_n) \Longrightarrow (S, D_h, \mu_h) \tag{2.6}$$

in Gromov–Hausdorff–Prokhorov law. Here d_n gives each original edge length one. Replace edges by cables of length one and linearly interpolate each selected half-edge step in one unit of discrete time. For every *fixed* integer $r \geq 1$, the source claims

$$\left(\overline{M}_n, a_n d_n, \mu_n; (\overline{X}^{n,j}_{nt})_{t\geq 0, 1\leq j\leq r}\right) \Longrightarrow \left(S, D_h, \mu_h; (B^{h,j}_t)_{t\geq 0, 1\leq j\leq r}\right). \tag{2.7}$$

Given the map the starts and walks are independent with initial law μ_n; given h the limiting motions are independent with initial law μ_h. Convergence includes locally uniform convergence of curves in common ambient metric spaces, together with Hausdorff convergence of spaces and Prokhorov convergence of masses. It does not assert convergence for $r = r_n \to \infty$, arbitrary starting vertices, or arbitrary maps satisfying only (2.6).

3 Imported Inputs and Source Proof Map

3.1 What is assumed beyond GHP convergence

From every subsequence, the source's input proposition takes a further subsequence and a common representation coupling. On one probability-one event it asserts all of the following simultaneously.

H1. There are conformal vertex coordinates $z_n : V_n \to S^2$ with $(z_n)_*\mu_n \Rightarrow \mu_h$, vanishing vertex distortion $\delta_n = \sup_{u,v} |a_n d_n(u,v) - D_h(z_n(u), z_n(v))|$, vanishing D_h-density error, and vanishing round diameter of all embedded flags. The full incidence flags matter at loops and repeated incidences.

H2. A corner tour lists $v_n(i)$, $0 \le i < 2n$, repeating vertex v exactly d_v times. A continuous $\eta : [0,1] \to S^2$ satisfies $\eta_* dt = \mu_h$, almost-everywhere singleton fibers, and uniform tracking of $z_n(v_n(\lfloor 2nt \rfloor))$ by $\eta(t)$.

H3. On $H = L^2([0,1], dt)$ set

$$(J_n u)(t) = u(v_n(\lfloor 2nt \rfloor)), \quad Jf = f \circ \eta, \quad K_n = J_n (2G_n)^\dagger J_n^*, \quad K = J(2G)^\dagger J^*.$$

The J_n are isometries and J is unitary. The dagger is inverse on mean-zero functions and zero on constants. K_n is also zero on $(\operatorname{ran} J_n)^\perp$. The input is $\|K_n - K\|_{\mathrm{HS}} \to 0$; constants are the only zero mode of G.

H4. For every buffered chart test $\operatorname{supp} \chi \subset V \Subset W \Subset U$, with $0 \le \chi \le 1$ smooth and U within an ordinary conformal chart, there are local recoveries $\widetilde{\chi}_n$ uniformly approximating χ near $\overline{W}$ and satisfying

$$\limsup_n \sum_{e : z_n(e) \in W} (d_e \widetilde{\chi}_n)^2 \le \int_U |\nabla \chi|^2 \, \mathrm{dvol}_0.$$

The assertion is simultaneous on the coupled outcome, including outcome-dependent buffered tests; edges incident on the compact support eventually lie wholly in W. The coefficient is one.

The source obtains H1–H2 from companion Proposition 2.4, H4 from Lemma 5.1 and Theorem 5.2, the form identification from Proposition 8.3, and H3 from Section 9, Lemma 9.1 and its inverse-HS equation. The relevant source files are respectively `02-model-geometry`, `06-conductivity`, `08-continuum`, and `09-spectrum`. Their filename prefixes do not coincide with every printed section number.

The simultaneous recovery quantifier is substantive. Having, for each fixed deterministic cutoff, a separate probability-one event does not automatically license arbitrary cutoffs selected after seeing the environment. The source explicitly imports a joint extraction of countably many constrained-energy infima for this purpose. We preserve that hypothesis; finite algebra does not establish it.

Spatial scale is also separate from time. The source uses the unique zero $\mathfrak{m}(Y)$ of $r \mapsto \mathbb{E} \arctan(Y - r)$ to define a deterministic diameter normalization $\widehat{a}_n$. Weak convergence and boundedness of arctangent imply continuity of this center, while $\mathfrak{m}(Y + c) = \mathfrak{m}(Y) + c$. Since GHP convergence implies convergence of scaled diameters to a finite positive diameter, these facts yield $\widehat{a}_n / a_n \to 1$. They permit the same geometric coupling for any a_n in (2.6); they do not calibrate conductivity or the walk clock.

3.2 Centered inverses to positive-time laws

Write Π for projection onto constants in H and $\psi_s(y) = e^{-s/y}$ for $y > 0$, $\psi_s(0) = 0$. Then the source proves

$$J_n e^{-sG_n} J_n^* = \Pi + \psi_s(2K_n), \quad s > 0, \tag{3.1}$$

$$J_n \alpha(\alpha + G_n)^{-1} J_n^* = \Pi + r_\alpha(2K_n), \quad r_\alpha(y) = \frac{\alpha y}{1 + \alpha y}, \quad \alpha > 0. \tag{3.2}$$

H3 implies norm convergence by continuous functional calculus on a common bounded spectral interval. Both formulas vanish on the discrete orthogonal complement and act as identity on constants. At $s = 0$ the left side is $J_n J_n^*$, a finite-rank projection, so convergence in operator norm

to the identity on infinite-dimensional H is impossible. This distinction is precisely the kind of uniformity issue emphasized in [4], although its indefinite pollution example is not an allegation of pollution for these positive graph generators.

For continuous $f_0, \dots, f_k$ on S^2 and strictly increasing times, stationarity gives a product of multiplications by $J_n(f_i \circ z_n)$ and the semigroups in (3.1), paired against 1 at both ends. H2 gives uniform convergence of those multipliers. Passing each factor to the limit identifies all finite-dimensional distributions. Repeated times are merged. This proves distributions at observation times, not control of spatial excursions between them.

3.3 Cutoff recovery to tightness

For $f = 1 - \chi$ in H4, subtract a uniform error bound $e_n \downarrow 0$ from $\widetilde{\chi}_n$, clip to $[0, 1]$ inside W, and set it to zero outside. A positive clipped value can occur only where $\chi > 0$. Thus every edge with a nonzero difference is inside the recovery buffer. The clipping map is a contraction, so the global $w_n = 1 - \chi_n$ obeys

$$\|w_n - f \circ z_n\|_\infty \to 0, \qquad \limsup_n \mathcal{E}_n(w_n) \le \mathcal{E}_h(f). \tag{3.3}$$

This is the source's uniform cutoff lemma, including the half factor.

For a stationary reversible finite chain of generator $-A$, source Lemma `lem:stationary-maximal` proves

$$\mathbb{E}_\pi \sup_{t \le T} |v(Y_t)|^2 \le 2\|v\|^2_{L^2(\pi)} + 40T\langle v, Av\rangle_\pi. \tag{3.4}$$

Its forward martingale is $N_t = v(Y_t) - v(Y_0) + \int_0^t Av(Y_s)\,\mathrm{d}s$, with $\mathbb{E}|N_T|^2 = 2T\langle v, Av\rangle_\pi$. Stationarity and reversibility give the same law backward. On a countable dense set of deterministic times the two identities subtract to eliminate the drift; right continuity then controls all times. Doob's L^2 bound yields the displayed constants. This is the finite jump-chain version of the classical Lyons–Zheng method [9], not a new Ram inequality.

For fixed $\alpha > 0$ put $g_n = f \circ z_n$ and $u_{n,\alpha} = \alpha(\alpha + G_n)^{-1} g_n$, $u_\alpha = \alpha(\alpha + G)^{-1} f$. Equation (3.2) and the resolvent equation give strong convergence of both $J_n u_{n,\alpha}$ and $J_n G_n u_{n,\alpha}$. Expanding $\mathcal{E}_n(w_n - u_{n,\alpha})$ and using (3.3) yields

$$\limsup_n \big[\|w_n - u_{n,\alpha}\|^2_{\mu_n} + \mathcal{E}_n(w_n - u_{n,\alpha})\big] \le \|f - u_\alpha\|^2_{\mu_h} + \mathcal{E}_h(f - u_\alpha).$$

The right side equals $\int (1+\lambda)[\lambda/(\alpha+\lambda)]^2\,\mathrm{d}\nu_f(\lambda)$. It tends to zero because f belongs to the form domain. Combining with (3.4) gives

$$\lim_{\alpha \to \infty} \limsup_n \mathbb{E}_{\mu_n} \sup_{t \le T} |u_{n,\alpha}(Y^n_t) - g_n(Y^n_t)| = 0. \tag{3.5}$$

Markov resolvents preserve $[0, 1]$, so $0 \le u_{n,\alpha} \le 1$ and $\|G_n u_{n,\alpha}\|_\infty \le \alpha$.

Choose finitely many buffered cutoffs f_i separating points at round distance at least $d > 0$, and an inner-disk cover on which each matching f_i is zero. For stopping times $0 \le \tau \le \sigma \le T$ with $\sigma - \tau \le s$, the first applicable disk index is $\mathcal{F}_\tau$-measurable. Optional sampling, weighted by these events, gives

$$\mathbb{P}\{\rho(z_n Y^n_\tau, z_n Y^n_\sigma) \ge d\} \le \alpha s + 2 \sum_i \mathbb{E} \sup_{t \le T} |f_i(z_n Y^n_t) - u^{(i)}_{n,\alpha}(Y^n_t)|. \tag{3.6}$$

First take $\limsup_n$, then $s \downarrow 0$ at fixed α, and finally $\alpha \to \infty$. Aldous's criterion [10] gives tightness in $D([0, T], S^2)$ with J_1 topology. The flag mesh makes all jump sizes vanish, so every J_1 limit is continuous. The finite-dimensional laws now identify the limiting path law. Interchanging these limits, omitting stationarity in (3.4), or replacing H4 by eigenvalue convergence alone would not reproduce this proof.

3.4 Metric transfer, attempt clock, and random environments

The source attaches each vertex to its coordinate image with bridge length $c_n = \delta_n/2 + a_n$. Triangle inequalities with $2c_n \geq \delta_n$ show that this construction shortens neither the cable metric nor D_h. The completion of the union is compact: the tail of the graphs lies arbitrarily close to the compact sphere. Its Hausdorff error is at most $\max(a_n + c_n, \varepsilon_n + c_n)$, and its mass error tends to zero. No extension of z_n to cable interiors is required. Uniform equivalence of ρ and D_h on the compact sphere transfers the path convergence.

Let $e_{n,k}$ be independent exponential variables of mean $1/n$, independent of the selected half-edges, and let $S_{n,k}$ be their partial sums. The process $Y_t^n = X^n_{N_n(t)}$ has generator $n(P_n - I)$, including invisible loop attempts. For $\Delta_{n,T} = \max_{k \leq \lfloor nT \rfloor} |S_{n,k} - k/n|$, Doob's inequality gives

$$\mathbb{E}\Delta^2_{n,T} \leq \frac{4\lfloor nT \rfloor}{n^2}, \qquad \sup_{t \leq T} d_Z(\overline{X}^n_{nt}, Y^n_t) \leq a_n + \omega_{T+1}(Y^n, \Delta_{n,T} + 1/n) \tag{3.7}$$

on the event that the argument of ω is at most one. The continuous limit controls this modulus, hence the difference vanishes in probability. Counting only genuine vertex changes would not justify this equation in a graph with loops.

For each fixed environment outcome and fixed r, products of the conditional path laws converge. Bounded convergence integrates these measurable sampling kernels over environments. The ambient space and recovery functions may be pointwise witnesses; it is the original conditional expectations that must be measurable. The subsequence principle yields the full distributional assertion. This is a reconstruction of the source's local proof, not an independent certification of H1–H4 or an assertion of quenched convergence on a preassigned almost-sure coupling of the entire original sequence.

4 Physical Ram Representation

4.1 Synthesis carries both energy and mass

Fix one finite graph and an injective synthesis matrix $Q : \mathbb{R}^m \to \mathbb{R}^{V_n}$, whose columns are selected Ram atoms or moments in vertex values. No invariance under G_n is assumed. Set

$$H_Q = Q^T M_n^{\mathrm{mass}} Q > 0, \quad F_Q = \tfrac{1}{2} Q^T C_n Q, \quad B_Q = H_Q^{-1} F_Q, \quad \mathcal{P}_Q = Q H_Q^{-1} Q^T M_n^{\mathrm{mass}}. \tag{4.1}$$

Then $\|Qc\|^2_{\mu_n} = c^T H_Q c$ and $\mathcal{E}_n(Qc, Qc) = c^T F_Q c$. The compression satisfies $H_Q B_Q = B_Q^T H_Q$, and is similar to the symmetric positive matrix $H_Q^{-1/2} F_Q H_Q^{-1/2}$. These statements follow by multiplication, using (2.3); they specialize the physical Gram and metric-symmetry propositions of [3].

An invertible analysis map $y = Tc$ changes both forms by congruence:

$$H_y = T^{-T} H_Q T^{-1}, \quad F_y = T^{-T} F_Q T^{-1}, \quad B_y = T B_Q T^{-1}.$$

For a physical source f, the weak load is $\ell_f = Q^T M_n^{\mathrm{mass}} f$, transforming as $T^{-T}\ell_f$. Using F_Q as the time generator, replacing H_Q by the identity, or retaining a load in the old coordinates generally changes the physical problem. A square invertible Q is merely an exact change of coordinates; compression is an approximation only if $m < |V_n|$.

4.2 Domains, complete residuals, and what is controlled

For any $c \in \mathbb{R}^m$ and real λ, orthogonality in $L^2(\mu_n)$ gives the inherited exact decomposition

$$\|(G_n - \lambda)Qc\|^2_{\mu_n} = \|(B_Q - \lambda)c\|^2_{H_Q} + \|(I - \mathcal{P}_Q)G_n Qc\|^2_{\mu_n}. \tag{4.2}$$

For a computed resolvent vector $\widehat{u}$ and exact $u = \alpha(\alpha + G_n)^{-1}g$, let $r = \alpha g - (\alpha + G_n)\widehat{u}$. Positivity gives the elementary physical control

$$\|u - \widehat{u}\|_{\mu_n} \le \alpha^{-1}\|r\|_{\mu_n}. \tag{4.3}$$

This is classical resolvent stability applied to the full residual. It is included as a representation check, distinct from the perturbation estimate in Section 5. In the continuum the same estimate requires $\widehat{u} \in D(G)$, not just $\mathcal{D}_h$. The omitted component in (4.2) cannot be dropped merely because a reduced linear system was solved exactly. Relative Gram inequalities $aH_Q \preceq \widehat{H}_Q \preceq bH_Q$ convert norms; entrywise-small mass errors alone do not.

4.3 Gauge compatibility and the static/dynamic boundary

On a connected graph write μ for the column vector of masses. The gauge kernel from [5], specialized to C_n, is

$$C_\mu^\dagger = (I - \mathbf{1}\mu^T)C_n^\dagger(I - \mu\mathbf{1}^T). \tag{4.4}$$

For a mean-zero function g, the dual charge $b = M_n^{\mathrm{mass}}g$ has $\mathbf{1}^T b = 0$. The solution of $(2G_n)u = g$, $\mu^T u = 0$, is $u = C_\mu^\dagger b$. Thus the centered inverse on functions includes the mass multiplication; it is not the Euclidean pseudoinverse of G_n read entrywise. Constants must also be restored in (3.1)–(3.2).

The exact elimination pairing in [5] is a static zero-sum load statement. If charges are supported on retained vertices, their $b^T C_n^\dagger d$ pairing is preserved by the Schur complement of C_n. This does not imply preservation of the dynamic generator, holding times, corner mass, or interpolated path law. The present author does not promote static Kron reduction to a walk-limit theorem. The next section instead compares actual weighted vertex walks.

5 Uniformly Small Conductance Perturbations

This local extension first compares walks on the same finite graph; only afterward does it invoke H1–H4 to transfer scaling limits. It is a classical variational perturbation argument with both physical mass and energy, not a result attributed to the pinned source.

5.1 Mass, energy, and two different clocks

Fix any connected finite multigraph with $n \ge 1$ original edges and positive weights c_e on every unoriented edge, including loops. Put $C_{\mathrm{tot}} = \sum_e c_e$ and $d_c(v) = \sum_{\text{half-edges at } v} c_e$, counting loops twice. The weighted half-edge walk P_c has reversible mass $\mu_c(v) = d_c(v)/(2C_{\mathrm{tot}})$. Each half-edge has directed stationary flux $c_e/(2C_{\mathrm{tot}})$. At clock C_{tot} the energy is $\frac{1}{2}\sum_e c_e d_e u\, d_e v$; at the required fixed clock it is

$$G_c = n(I - P_c), \qquad \mathcal{E}_c(u,v) = \frac{n}{2C_{\mathrm{tot}}}\sum_e c_e d_e u\, d_e v. \tag{5.1}$$

Subscript 0 denotes unit weights and their *half*-energy form, not the companion's raw convention. Common scaling of all weights leaves P_c, μ_c, G_c unchanged at the n-clock, but scales the C_{tot}-clock operator. Loops change mass and holding probabilities despite their zero energy differences.

If $0 \le \epsilon < 1$ and $1 - \epsilon \le c_e \le 1 + \epsilon$ on every edge, define

$$a = \frac{1-\epsilon}{1+\epsilon}, \quad b = a^{-1}, \quad \delta = b - 1 = \frac{2\epsilon}{1-\epsilon}, \quad \rho = \mu_c/\mu_0. \tag{5.2}$$

Comparing degrees, total weights, and each coefficient nc_e/C_{tot} gives

$$a \le \rho(v) \le b, \qquad a\mathcal{E}_0(u) \le \mathcal{E}_c(u) \le b\mathcal{E}_0(u). \tag{5.3}$$

Edgewise Cauchy–Schwarz and $|\rho - 1| \le \delta$ give

$$\begin{aligned} |(\mathcal{E}_c - \mathcal{E}_0)(u,v)| &\le \delta\sqrt{\mathcal{E}_0(u)\mathcal{E}_0(v)}, \\ |\langle u,v\rangle_c - \langle u,v\rangle_0| &\le \delta\|u\|_0\|v\|_0. \end{aligned} \tag{5.4}$$

Here $\|u\|_0$ means the baseline $L^2(\mu_0)$ norm.

Theorem 5.1 (Uniform positive-resolvent perturbation)**.** *For every such graph, every admissible conductance assignment, and every $\alpha > 0$, let $R_i(\alpha) = \alpha(\alpha + G_i)^{-1}$, $i = 0, c$, act on the same vertex functions. Then*

$$\|R_c(\alpha) - R_0(\alpha)\|_{0\to 0} \le D_\epsilon := \frac{\epsilon}{1-\epsilon}. \tag{5.5}$$

The norm is the baseline physical operator norm. The bound is uniform in n, α, degrees, loops, multiplicities, and spectral gap. No inverse statement at $\alpha = 0$ or sharpness claim is made.

Proof. Multiply both the weighted mass and energy by C_{tot}/n. The resulting auxiliary mass and energy are

$$\widetilde{m}_c(v) = \frac{d_c(v)}{2n}, \qquad \widetilde{\mathcal{E}}_c(u,v) = \tfrac{1}{2}\sum_e c_e d_e u\, d_e v. \tag{5.6}$$

This common scalar changes neither G_c nor its resolvent equation: $\langle u, G_c v\rangle_{\widetilde{m}_c} = \widetilde{\mathcal{E}}_c(u,v)$. The auxiliary total mass is C_{tot}/n, not necessarily one. It is used only in the variational proof; the stationary probability of the walk remains μ_c throughout the path theorem.

Now $\widetilde{\rho} = \widetilde{m}_c/\mu_0$ lies directly in $[1-\epsilon, 1+\epsilon]$, and the same bounds compare $\widetilde{\mathcal{E}}_c$ with $\mathcal{E}_0$. Both bilinear differences have constant ϵ instead of δ in (5.4). For any f put $u_i = R_i(\alpha)f$, $e = u_c - u_0$, $A_0(u,v) = \alpha\langle u,v\rangle_0 + \mathcal{E}_0(u,v)$, and $\widetilde{A}_c(u,v) = \alpha\langle u,v\rangle_{\widetilde{m}_c} + \widetilde{\mathcal{E}}_c(u,v)$. Subtracting the variational equations gives exactly

$$\widetilde{A}_c(e,v) = \alpha\langle(\widetilde{\rho}-1)(f-u_0), v\rangle_0 - (\widetilde{\mathcal{E}}_c - \mathcal{E}_0)(u_0, v). \tag{5.7}$$

Let ν_f be the spectral measure of G_0 at f. Spectral calculus gives the joint identity

$$\begin{aligned} \alpha\|f-u_0\|_0^2 + \mathcal{E}_0(u_0) &= \int_{[0,\infty)} \frac{\alpha\lambda^2 + \alpha^2\lambda}{(\alpha+\lambda)^2}\,\mathrm{d}\nu_f(\lambda) \\ &= \int_{[0,\infty)} \frac{\alpha\lambda}{\alpha+\lambda}\,\mathrm{d}\nu_f(\lambda) \le \alpha\|f\|_0^2. \end{aligned} \tag{5.8}$$

Set $v = e$ in (5.7) and use the auxiliary ϵ-bilinear bounds. Cauchy–Schwarz in the paired mass and energy components yields

$$\begin{aligned} |\widetilde{A}_c(e,e)| &\le \epsilon\left[\alpha\|f-u_0\|_0\|e\|_0 + \sqrt{\mathcal{E}_0(u_0)\mathcal{E}_0(e)}\right] \\ &\le \epsilon\sqrt{\alpha\|f-u_0\|_0^2 + \mathcal{E}_0(u_0)}\sqrt{A_0(e,e)} \\ &\le \epsilon\sqrt{\alpha}\|f\|_0\sqrt{A_0(e,e)}. \end{aligned}$$

Also $\widetilde{A}_c(e,e) \ge (1-\epsilon)A_0(e,e)$ and $\sqrt{\alpha}\|e\|_0 \le \sqrt{A_0(e,e)}$. Dividing when the last quantity is nonzero gives $\|e\|_0 \le [\epsilon/(1-\epsilon)]\|f\|_0$; the zero case is immediate. A one-vertex loop graph is included. No degree or gap estimate entered the proof. □

For comparison, applying the same joint budget directly to the normalized probability mass and energy, without (5.6), gives $\delta/a = 2\epsilon(1+\epsilon)/(1-\epsilon)^2$. Using in addition the separate estimates $\|f-u_0\|_0 \le \|f\|_0$ and $\mathcal{E}_0(u_0) \le (\alpha/4)\|f\|_0^2$ instead of the joint budget gives the still valid but weaker constant

$$D_\epsilon^{\mathrm{old}} = \frac{\sqrt{5}\,\epsilon(1+\epsilon)}{(1-\epsilon)^2}. \tag{5.9}$$

The two improvements use the joint identity (5.8) and common mass/energy rescaling, respectively. They are not a sharpness claim for the perturbation problem. The finite tests were executed for $D_\epsilon^{\mathrm{old}}$ only. They are not relabeled as verification of the smaller bound. Connectedness ensures all degrees are positive; a disconnected variant must remove isolated zero-degree vertices or specify a separate positive-mass transition convention.

Corollary 5.2 (Paired same-graph semigroups)**.** *Set $S_i(t) = e^{-tG_i}$ and $\eta_k = \sup_{x\ge 0}[(1+x/k)^{-k} - e^{-x}]$. For every $t > 0$ and integer $k \ge 1$,*

$$\|S_c(t) - S_0(t)\|_{0\to 0} \le (1+b)\eta_k + kbD_\epsilon. \tag{5.10}$$

Consequently, along any paired graph sequence with $\epsilon_n \to 0$, this difference tends to zero uniformly for $t > 0$.

Proof. The scalar difference is nonnegative and $\eta_k \to 0$. On compact x-intervals convergence is uniform; for $x \ge A$ the difference is at most $(1+A/k)^{-k}$. Send $k \to \infty$ and then $A \to \infty$. The spectral theorem bounds $S_i(t) - R_i(k/t)^k$ by η_k in its own Hilbert norm. Conversion from the weighted to the baseline operator norm costs $\sqrt{b/a} = b$. Every power $R_c(\alpha)^j$ has baseline norm at most b, not b^j, since it is a contraction in the weighted norm. Telescoping the k resolvent factors therefore costs at most kbD_ϵ. This proves (5.10). Fix k, take $\epsilon_n \to 0$, then $k \to \infty$. Both operators have the same finite identity at time zero; the conclusion is not time-zero norm convergence from finite embedded spaces to an infinite space. □

5.2 Conditional weighted path extension

Theorem 5.3 (Vanishing conductance perturbations)**.** *Assume the complete source package H1–H4, with its original spatial metric, tour, and half-energy LBM. Give every original edge weights $1 - \epsilon_n \le c_{n,e} \le 1 + \epsilon_n$, where the bounds are deterministic and $\epsilon_n \to 0$. The weights may be measurable functions of the map or use auxiliary randomness conditional on it. Given this weighted environment, start any fixed r independent weighted half-edge walks independently in μ_n^c. The limit* (2.7) *holds for their cable paths with the* same exact n-clock*. The discrete mass in the decorated object may be μ_n^c or μ_n^0. The conclusion also holds with starts in μ_n^0, or any law total-variation-close to μ_n^c on the coupling. All edge lengths and the spatial scale are unchanged.*

Proof. Fix a good source outcome and its extracted sequence. Put $\rho_n = \mu_n^c/\mu_n^0$, $r_n = J_n\rho_n$, and let T_n multiply corner-time functions by $\sqrt{r_n}$. Then $T_n, T_n^{-1} \to I$ in operator norm and preserve $\operatorname{ran} J_n$ and its orthogonal complement. The exact weighted isometry is

$$U_n f = J_n(\sqrt{\rho_n} f) = T_n J_n f, \qquad U_n^* = \rho_n^{-1/2} J_n^*. \tag{5.11}$$

In the adjoint formula J_n^* is the baseline adjoint. Indeed $\|U_n f\|_H^2 = \mu_n^c(|f|^2)$. The weighted constant maps to $e_n = \sqrt{r_n}$, of norm one, and $e_n \to 1$ uniformly. Its projection is $e_n \otimes e_n$, not Π.

The nonnegative self-adjoint contraction on H representing the weighted resolvent is

$$\widetilde{R}_n^c(\alpha) = U_n R_n^c(\alpha) U_n^* = T_n[J_n R_n^c(\alpha) J_n^*] T_n^{-1}.$$

Theorem 5.1, $T_n^{\pm 1} \to I$, and the source's baseline convergence imply norm convergence to $J\alpha(\alpha + G)^{-1}J^*$ for each fixed $\alpha > 0$. For $s > 0$ apply the continuous function on $[0, 1]$

$$F_{s,\alpha}(y) = \begin{cases} \exp[-s\alpha(1/y - 1)], & y > 0, \\ 0, & y = 0. \end{cases}$$

Its values at zero and one handle the discrete complement and weighted constants, respectively. Continuous functional calculus gives $U_n e^{-sG_n^c} U_n^* \to Je^{-sG}J^*$ in norm. No convergence of weighted centered inverses is inferred. Multiplication by $g_n = f \circ z_n$ is represented by multiplication by $J_n g_n$, *not* by $U_n g_n$, because $U_n(g_n v) = M_{J_n g_n} U_n v$. Thus stationary finite-dimensional laws converge, using e_n at both ends of the product, $e_n \to 1$, uniform tracking, and positive-time semigroup convergence.

Retain the original cutoff recoveries w_n for $f = 1 - \chi$. Their energies are bounded and $\mathcal{E}_n^c(w_n) \le b_n \mathcal{E}_n^0(w_n)$ with $b_n \to 1$, hence $\limsup_n \mathcal{E}_n^c(w_n) \le \mathcal{E}_h(f)$ and $U_n w_n \to Jf$. For fixed α, set $u_n^c = R_n^c(\alpha) g_n$. The resolvent limit gives $U_n u_n^c \to J u_\alpha$ and its equation gives $U_n G_n^c u_n^c \to JGu_\alpha$. Expanding the energy difference in the actual weighted mass yields

$$\limsup_n [\|w_n - u_n^c\|_{\mu_n^c}^2 + \mathcal{E}_n^c(w_n - u_n^c)] \le \|f - u_\alpha\|_{\mu_h}^2 + \mathcal{E}_h(f - u_\alpha).$$

The right side tends to zero by the form-domain spectral argument. The weighted chain is reversible and stationary in μ_n^c, so (3.4) applies with $A = G_n^c$ and proves its version of (3.5). Weighted Markov resolvents still satisfy $0 \le u_n^c \le 1$ and $\|G_n^c u_n^c\|_\infty \le \alpha$. The same buffered cutoff cover and optional-sampling argument give (3.6), with unchanged limit order n, s, α. Compact containment and vanishing jump size use the original geometry. Tightness and the finite-dimensional laws identify LBM.

The mass comparison gives $\mathrm{TV}(\mu_n^c, \mu_n^0) = \frac{1}{2}\mu_n^0(|\rho_n - 1|) \to 0$. Thus either mass has the same metric-measure limit, and the original cable ambient construction applies. Poissonizing weighted half-edge choices at attempt rate n gives precisely $n(P_n^c - I)$, including loop attempts. The exponential clock deviation in (3.7) is unchanged. Weighted tightness supplies its required modulus, while movement per cable step is still at most a_n. This removes the clock at exact acceleration n.

For a fixed Markov path kernel, total variation of path laws is at most that of the initial measures. The claim for μ_n^0 and other close starts follows; for r independent copies the difference is at most r times the individual distance. Products are taken conditional on the weighted environment before averaging. On each good baseline outcome the argument holds for every sequence of weights within the deterministic bounds. Extra weight randomness may be sampled after coupling the original maps and needs no separate scaling limit. Measurable sampling kernels, bounded convergence, and the source's subsequence principle finish the joint claim. □

No arbitrary-conductance invariance principle is proved. Fixed nonvanishing distortions, weighted cable metrics, arbitrary vertex starts, and increasing r are outside the statement. In particular, (5.5) alone is not a path theorem: recovery, maximal estimates, mesh control, and the attempt-clock proof are essential. It supplies no numerical convergence rate in n to the continuum.

For weighted Ram synthesis, replace (4.1) by

$$H_Q^c = Q^T \operatorname{diag}(\mu_c) Q, \qquad F_Q^c = \frac{n}{2C_{\mathrm{tot}}} (D_n Q)^T \operatorname{diag}(c)(D_n Q).$$

The physical symmetry and full residual account remain exact. A positive compressed form need not define a Markov chain in an arbitrary coefficient basis; the path theorem concerns vertex walks.

6 A Conditional Initial-Density Extension

This section adds a bounded result not stated in the source's stationary theorem. Its engine is the classical change-of-initial-law identity. It does not rerun or strengthen the deep electrical proof.

Fix any outcome and extracted sequence satisfying H1–H4 and the source's stationary cable-path conclusion in a common compact ambient space Z. Write $W_n(t) = \overline{X}^n_{nt}$ and $W(t) = B^h_t$. Let P_n be the stationary law of W_n and P that of W. The following criterion is in the common coordinate, and is not merely weak convergence of starting measures.

Theorem 6.1 (L^1-compatible starts)**.** *Suppose $q_n : V_n \to [0,\infty)$ and $q \in L^1(S^2, \mu_h)$ are probability densities: $\mu_n(q_n) = \mu_h(q) = 1$. Extend J_n, J to their natural L^1 isometries. Assume*

$$\|J_n q_n - Jq\|_{L^1([0,1])} \longrightarrow 0. \tag{6.1}$$

Start the same walks instead from $q_n\mu_n$, retaining their transition rules and exact n-clock. Then their cable-path laws converge in $C([0,\infty), Z)$ to LBM started from $q\mu_h$. No boundedness of q_n or q is needed. The statement holds for any fixed finite product of independently sampled walks with possibly different densities satisfying (6.1)*.*

Proof. For every bounded measurable path functional F, disintegration at time zero gives

$$\mathbb{E}_{q_n\mu_n} F(W_n) = \mathbb{E}_{\mu_n}[q_n(W_n(0))F(W_n)]. \tag{6.2}$$

The corresponding identity holds for $q\mu_h$ in the limit. It changes only the initial distribution, not the generator or the edge selection.

For any continuous $\varphi : S^2 \to \mathbb{R}$, uniform tour tracking and (6.1) imply

$$\limsup_n \|q_n - \varphi \circ z_n\|_{L^1(\mu_n)} \le \|q - \varphi\|_{L^1(\mu_h)}. \tag{6.3}$$

Continuous functions are dense in L^1 for the finite Radon measure μ_h on the compact sphere. Given $\epsilon > 0$, choose such a φ with right side below ϵ. Regard φ as a continuous function on the compact subspace (S, D_h) of Z and extend it continuously to Z. At a graph start, this extension evaluated at v differs uniformly from $\varphi(z_n(v))$ by $o(1)$, because the vertex bridges vanish. For bounded continuous F on path space, the functional $\varphi(W(0))F(W)$ is bounded continuous. Stationary path convergence therefore passes its expectation to the limit. The two errors from replacing q_n, q by φ are at most $2\epsilon\|F\|_\infty + o(1)$ by (6.2)–(6.3). Let $\epsilon \downarrow 0$. Convergence against all bounded continuous F is weak convergence of the tilted probability laws in the Polish path space. Apply this argument to each of finitely many densities; products of weakly convergent probability laws converge. Conditional independence is preserved by sampling each tilted law separately. □

Proposition 6.2 (Event transfer and tightness budget)**.** *For any path event A, any density q_n, and $R > 0$, define $\theta_n(R) = \int_{\{q_n > R\}} q_n \, \mathrm{d}\mu_n$. Then*

$$\mathbb{P}_{q_n\mu_n}(A) \le R\mathbb{P}_{\mu_n}(A) + \theta_n(R). \tag{6.4}$$

If the family $J_n q_n$ is uniformly integrable, tightness of the stationary laws transfers to the tilted laws. If additionally $\sup_n \|q_n\|_{L^p(\mu_n)} \le C$ for some $p > 1$, then

$$\mathbb{P}_{q_n\mu_n}(A) \le C\mathbb{P}_{\mu_n}(A)^{1-1/p}. \tag{6.5}$$

In particular these statements apply to the stopping-time events in (3.6) *and to failed attempt-clock comparisons, on their joint walk/clock probability spaces.*

Proof. Split the expectation of $q_n(W_n(0))\mathbf{1}_A$ at $q_n = R$ to get (6.4). Given an error tolerance, uniform integrability first chooses R with uniformly small tails, then stationary tightness chooses a compact set with sufficiently small complement probability. For the finite number of early indices not covered by an eventual bound, enlarge the compact set using tightness of each individual law. Hölder's inequality gives (6.5), since the initial marginal under the stationary law is exactly μ_n. Strong L^1 convergence in (6.1) implies uniform integrability on $[0, 1]$. The same Radon–Nikodym factor applies when independent exponential attempt clocks are adjoined. □

Uniform integrability transfers tightness, but does not alone identify which starting density occurs in the limit. Condition (6.1) provides that identification. Conversely the source's stationary maximal inequality is not being asserted under $q_n\mu_n$: it is the *whole path law* that is reweighted after the stationary estimate has been established.

Random-environment formulation. For an annealed extension, require that on every source representation subsequence the densities admit the same coupled L^1 limit q_h, almost surely, and that the tilted sampling kernels are measurable functions of the retained environments. The limiting density must define an intrinsic probability on the surface, or else the coordinate/mark data specifying it must be retained in the statement. Theorem 6.1 then gives pointwise convergence of conditional expectations; bounded convergence and the subsequence principle give the decorated distributional limit with initial law $q_h\mu_h$. No environment-uniform L^p bound is required for this argument. Without the measurability and coupling hypotheses, only the fixed-outcome theorem above has been proved.

7 Corner Moments and Concrete Classes of Starts

7.1 A positive, normalized realization of every L^1 density

The source's corner isometry has a useful additional consequence. Let $I_n(v)$ be the union of the d_v half-open intervals $[i/(2n), (i + 1)/(2n))$ assigned to v. Its Lebesgue measure is exactly $\mu_n(v)$. These sets form a measurable partition up to endpoints.

Proposition 7.1 (Corner-average density synthesis)**.** *For every probability density $q \in L^1(\mu_h)$ define*

$$q_n(v) = \frac{1}{\mu_n(v)} \int_{I_n(v)} q(\eta(t))\,\mathrm{d}t. \tag{7.1}$$

Then $q_n \geq 0$, $\mu_n(q_n) = 1$, and (6.1) holds. For every $p \in [1, \infty]$ for which $q \in L^p(\mu_h)$, $\|q_n\|_{L^p(\mu_n)} \leq \|q\|_{L^p(\mu_h)}$. These are positive mass-weighted Ram cell moments in corner time.

Proof. Let E_n be conditional expectation onto the partition $\{I_n(v)\}_{v\in V_n}$. It is positive, preserves integrals, and contracts every L^p, so $J_nq_n = E_nJq$ gives all conclusions except convergence. The partitions need not be nested and a single vertex can appear in widely separated tour intervals.

For continuous f on S^2, $J_n(f \circ z_n)$ is partition-measurable and uniformly tends to Jf. Hence

$$\|E_nJf - Jf\|_1 \leq 2\|J_n(f \circ z_n) - Jf\|_1 \longrightarrow 0.$$

By the unitary, measure-preserving identification of H2 and density of continuous functions in $L^1(\mu_h)$, approximate q by such an f. Contractivity gives

$$\|E_nJq - Jq\|_1 \leq 2\|q - f\|_{L^1(\mu_h)} + \|E_nJf - Jf\|_1.$$

Take $n \to \infty$, then $f \to q$. Jensen's inequality proves the L^p claim, including the usual essential-supremum interpretation. □

This is a conditional realization on the representation coupling, not an algorithm extracting the unknown limiting sphere from one finite map. It uses its actual tour and limit density, and makes no runtime claim. In finite coordinates $I_n(v)$ can be reordered without changing mass; arbitrary averaging in an unweighted vertex index is different.

7.2 Continuous tilts and region-conditioned starts

For a nonnegative continuous f with $\mu_h(f) > 0$, let

$$q_n(v) = \frac{f(z_n(v))}{\mu_n(f \circ z_n)}, \qquad q = \frac{f}{\mu_h(f)}. \tag{7.2}$$

The denominator converges to a positive number and uniform tracking gives even uniform convergence of the pullbacks, so Theorem 6.1 applies. This is a simple implementable start rule when the conformal coordinates are available.

Corollary 7.2 (Continuity-region starts)**.** *Let $D \subset S^2$ be Borel with $\mu_h(D) > 0$ and $\mu_h(\partial D) = 0$, for a fixed coupled outcome. Start uniformly with respect to corner mass conditioned on $z_n(v) \in D$. Then the conclusion of Theorem 6.1 holds with $q = \mathbf{1}_D/\mu_h(D)$, for all sufficiently large n.*

Proof. At every t whose image $\eta(t)$ is outside ∂D, uniform tour tracking makes membership in D eventually agree with that of the discrete vertex: such a point is in either the interior of D or the interior of its complement. The exceptional preimage has Lebesgue measure zero by $\eta_* dt = \mu_h$. Dominated convergence gives L^1 convergence of indicators and convergence of their normalizing masses to $\mu_h(D) > 0$. □

For a fixed center in a finite measure metric space, at most countably many radii have spheres of positive mass, since those spheres are disjoint. Thus almost every radius supplies a continuity ball. A shrinking ball is a different matter: its normalizing density can lose uniform integrability. Neither the corollary nor positivity of each finite starting mass proves point-start convergence.

7.3 A sharp logical boundary: rare starts

Consider a reversible two-state chain on $\{0, 1\}$, with rates $q(0, 1) = 1/n$ and $q(1, 0) = n$. Its stationary probabilities are $\pi_n(1) = 1/(n^2 + 1)$ and $\pi_n(0) = n^2/(n^2 + 1)$. In a fixed two-point metric, the stationary chance of any visit to state 1 by time T is at most $1/(n^2 + 1) + T/n$. Thus stationary paths converge to the constant zero path. But the density $r_n = (n^2 + 1)\mathbf{1}_{\{1\}}$ starts the process surely at 1, from which it exits in mean time $1/n$. These path laws are not tight at time zero in $D([0, T], \{0, 1\})$: any possible limit must start at 1, but its value at every fixed positive time is 0, contradicting right continuity. The density tails have integral one for every fixed cutoff once n is large, so uniform integrability fails. This is a general reversible-chain counterexample to an inference, *not* an FK-map counterexample or a challenge to the source's stationary theorem.

8 Exact Source-Specific Tests

8.1 A planar multigraph with both loops and parallel edges

Take three vertices $0, 1, 2$, one loop at 0, two edges between 0 and 1, and one edge between 1 and 2. Here $n = 4$, $(d_0, d_1, d_2) = (4, 3, 1)$, and

$$M^{\mathrm{mass}} = \mathrm{diag}(1/2, 3/8, 1/8), \qquad P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 2/3 & 0 & 1/3 \\ 0 & 1 & 0 \end{pmatrix}, \tag{8.1}$$

$$C = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 3 & -1 \\ 0 & -1 & 1 \end{pmatrix}, \qquad G = \begin{pmatrix} 2 & -2 & 0 \\ -8/3 & 4 & -4/3 \\ 0 & -4 & 4 \end{pmatrix}.$$

Thus $MG = C/2$ and $MP = P^T M$ exactly. The loop does not enter C, but it does enter M, P, n, and the rate of invisible attempts. Deleting the loop while pretending to keep the same walk would change these quantities. Each of the parallel edges contributes separately to C. Removing the FK decoration A from the dynamics is deliberate: the source never restricts transitions to occupied edges.

For the nonorthogonal trial columns $Q = [\mathbf{1}, (0, 1, 2)^T]$, the test script checks $H_Q B_Q = B_Q^T H_Q$, idempotence and physical symmetry of $\mathcal{P}_Q$, and (4.2) for all $c = (a, b)^T$ with $a, b \in \{-2, -1, 0, 1, 2\}$ and $\lambda \in \{-1, 0, 1, 2\}$. These are exact rational identities, not small floating-point discrepancies. The full-space equations and the general orthogonal-decomposition proof remain the evidence for all vectors, rather than an extrapolation from the finite list.

The centered raw inverse is computed as $K = (2G + \Pi)^{-1} - \Pi$, where $\Pi = \mathbf{1}\mu^T$. At $\alpha = 1/3, 1, 7$ the script checks

$$\alpha(\alpha I + G)^{-1} = \Pi + 2\alpha K(I + 2\alpha K)^{-1}.$$

The right-hand nonconstant term alone annihilates $\mathbf{1}$, so forgetting Π would fail even on a constant test function. These are checks of the source functional calculus, not empirical evidence for H3.

8.2 Initial-law and corner-moment checks

The density $q = (3/2, 1/2, 1/2)^T$ satisfies $\mu(q) = 1$ on (8.1). For every vertex word of lengths one through five, the exact path probability from $q\mu$ is checked to equal the stationary word probability times q at the initial vertex. Summing each length gives total probability one. This directly tests the finite identity (6.2), including words with repeated vertices due to the loop.

A corner list $[0, 0, 0, 0, 1, 1, 1, 2]$ is used to test (7.1). Positive rational corner data are normalized to mean one; averaging by vertex gives nonnegative density of exactly unit corner mass. Convex tail functionals $(x - R)_+$ at rational thresholds decrease under this conditional expectation, as Jensen's inequality predicts. This finite list is not claimed to be the FK exploration tour of a sampled map. It tests the multiplicity-dependent isometry used by the source, without inventing an FK sampler.

The rare-state example is checked at $n = 2, 3, 10, 100$ for detailed balance, initial-density normalization, and exact escape mean $1/n$. Its asymptotic conclusion follows from the proof in Section 7, not the four evaluations.

8.3 Weighted finite checks

A separate exact rational suite tests five connected graphs, including a single vertex with a loop, a single edge, and multigraphs with loops and parallel edges. Four weight cases per graph use

$\epsilon = 0, 1/10, 1/3, 3/4$, with all weights within $[1-\epsilon, 1+\epsilon]$. For each case and $\alpha = 1/7, 1, 13$, the script verifies the conservative bound (5.9), not the improved bound (5.5), by checking every principal minor of the rational symmetric matrix

$$(D_\epsilon^{\mathrm{old}})^2 M_0 - (R_c - R_0)^T M_0 (R_c - R_0)$$

is nonnegative. It also checks mass stationarity, weighted half-edge normalization, stiffness identity, constant resolvent mode, and invariance under common weight rescaling by seven at the n-clock. All 472 exact assertions pass. The smaller constant in (5.5) is proved analytically by (5.6)–(5.8); these finite checks do not certify it. Neither finite suite replaces a universal proof or tests the continuum hypotheses.

Reproducibility. The accompanying `evidence/author_checks.py` uses exact SymPy rationals and matrices. Its JSON output records the assertions, matrices, omitted residual, and script hash. There is no Monte Carlo scaling experiment, eigenvalue-only claim of path convergence, timing benchmark, or numerical claim about quantum maps. The author suite passes 495 exact assertions, in addition to the 472 weighted assertions. Its printed Gram matrix is $\left(\begin{smallmatrix} 1 & 5/8 \\ 5/8 & 7/8 \end{smallmatrix}\right)$, its compressed generator is $\left(\begin{smallmatrix} 0 & -60/31 \\ 0 & 96/31 \end{smallmatrix}\right)$, and the squared omitted residual of the second trial column is $2/93$.

9 Current Primary Literature and Attribution

Public primary sources were checked on October 8, 2026, using only public titles and general mathematical terms. Private Ram content was not included in any query or upload. All eleven local main/companion files in the recorded provenance scope match the public raw bytes at the supplied commit. The snapshot is not a local Git working tree, so this byte comparison, rather than a claimed local Git HEAD, is the provenance basis.

Berestycki and Gwynne [6], Theorems 1.2 and 1.4 and Remark 1.3, prove parametrized walk convergence on mated-CRT maps in conformal embeddings. Their deterministic scale is an annealed median-exit clock comparable to inverse cell area; the remark does not identify its limiting multiplicative constant. This is a different model and normalization from the source's finite FK sphere with exact edge-count clock. Their starting-point conclusions are stronger in their own setting; the conditional density extension here is not a new first result about nonstationary LBM limits in the literature.

Garban, Rhodes, and Vargas [7], Theorem 1.7 and Remark 1.8, give the Dirichlet-form identification and emphasize the reference measure's role. The displayed energy there is unhalved. The present source fixes its own half-energy convention explicitly; a citation to a Dirichlet-form formula without checking this factor cannot fix the n versus $2n$ clock. Their work also distinguishes the Dirichlet-form intrinsic pseudometric from the intended random geometry. We do not use it to infer metric GHP convergence.

The finite-volume inventory walk in Gwynne–Sun [8] encodes boundary lengths and is not the all-edge vertex walk of (2.1). The distinction prevents an encoding invariance principle from being substituted for the theorem being analyzed. Lyons–Zheng [9] and Aldous [10] are the classical sources for the stationary forward–backward method and stopping-time tightness criterion. Conditional expectation, change of measure, Hölder's inequality, Hilbert projections, and Schur elimination are also classical. The Ram role is disciplined source/metric/coordinate organization and the concrete initial-density realization, not ownership of these imported tools.

The public searches additionally checked recent FK/LBM terms and the public OpenAI repository. No comprehensive novelty review or absence theorem follows from these searches. The main source

and companion remain pinned preprints for purposes of this report; matching public bytes does not imply peer review or mathematical correctness.

10 Advantages, Limitations, and Further Work

The realized improvements are precise. The finite estimate (5.5) is uniform in graph, positive parameter, degrees, and spectral gap, and the conditional weighted theorem preserves the exact source clock under vanishing uniform perturbations. After the stationary source theorem is granted, compatible initial densities can be propagated through the *same* path theorem, metric coupling, and exact clock. The criterion is weak enough to include unbounded L^1 densities, and the positive corner-average construction proves that its class is nonempty and broad on each fixed representation coupling. The event bounds explain exactly how stationary tightness survives a change of initial distribution. This avoids falsely extending the stationary martingale proof itself to nonstationary starts.

The physical Ram account offers useful error diagnostics. It keeps mass with energy, handles constants and orthogonal complements, distinguishes form from operator domains, and forces full residuals to include components omitted by a trial bank. Corner multiplicities have a concrete role in both the norm and the new density realization. All of these benefits can be expressed in ordinary weighted linear algebra; changing coordinates alone is not a new diffusion theorem.

The principal costs and boundaries are equally concrete.

- H1–H4 include major unverified-in-this-report geometric and electrical statements. Neither the new density proof nor exact tests close any gap in their foundational derivation.
- Theorem 6.1 grants no arbitrary point-start result. Densities concentrating on tiny corner sets may lose uniform integrability; the two-state example explains the missing implication.
- The corner realization uses the limiting tour and density on a coupling. Without additional information it is an existence statement, not an efficient finite-map algorithm.
- The source supplies qualitative convergence, not numerical rates for inverse-HS error, recovery error, flag mesh, or metric distortion. The event estimates expose where rates would enter but supply none.
- A finite nonorthogonal bank may be ill-conditioned. Positivity of its Gram matrix does not bound its condition number or eliminate its omitted residual. No speedup or stability gain has been measured.

Future work could establish start-density convergence from observables of the finite decorated map without referencing the limiting tour; derive uniform quantitative cutoff-recovery rates; or combine actual pointwise heat-kernel control with exceptional-set estimates to approach specified vertex starts. These are separate projects, not conclusions. For increasing numbers of walks one would also need quantitative product-error control. None of these proposed improvements is counted in the contribution score.

The conservative assessment is therefore $C2/5$ at relevance $L4/5$. The weighted and initial-density theorems genuinely enlarge the conditional scope, but their methods are classical and the essential geometric and electrical inputs are inherited. No global FK–LQG proof, new unit-conductivity theorem, or important novelty/speed claim has been demonstrated here.

A Read Scope and Verification Boundary

The evidence directory records local SHA-256 values, public raw comparisons, public literature URLs and read locations, executable exact tests, native compiler attempts, physical build diagnostics, and final visual inspection. A public file hash records the entire file's bytes, not a claim that every line was read. Main-source files were read in full; companion scope is limited to the passages listed below.

- Companion `02-model-geometry.tex`, lines 192–277: the actual common geometric extraction statement, through most of its chart clause; no full geometry proof was audited here.
- Companion `06-conductivity.tex`, lines 77–108 and 198–231: recovery and unit-conductivity statements; no claim to have verified the long conductivity proof.
- Companion `08-continuum.tex`, lines 373–456: domain/inverse proposition and initial proof, including the trace and time-change explanation.
- Companion `09-spectrum.tex`, lines 1–78 and 448–479: corner-space setup, unitary identification statement, four-point kernel limit and inverse-HS conclusion, not the complete compactness argument.
- Private `ch01d-physical-metrics.tex` and `ch01f-spectral-limits.tex`: full text read, including their actual proofs and warnings; figure image files were not inspected.
- Private `TR-meta5-math.tex`: lines 115–196 and 451–524, covering graph energy, explicit general gauge, invariant pairing, transported forms/loads, static elimination and the start of the Laplace-moment discussion. No whole-report read is claimed.

An independent operator-to-path audit returned a bounded conditional pass of the source's local implication from H1–H4, with no blocking flaw found. It separately checked the weighted path transfer while taking the finite perturbation estimate as an input. This is not a certification of the spectral companion or the foundational geometry. Its detailed read scope and separate finite boundary examples are preserved in the evidence directory. The report's reconstruction and its local extensions remain distinct from that review.

References

[1] OpenAI, *Random Walks on Critical FK–Ising Maps and Liouville Brownian Motion*, October 5, 2026, pinned source, sections `inputs`, `operators`, `tightness`, `paths`. Public pinned source directory.

[2] OpenAI, *Spectral convergence for critical FK–Ising planar maps*, October 5, 2026, same pin, selected actual passages specified in Appendix A. Public pinned companion directory.

[3] M. Subbarao, *Ram Calculus*, Volume 21, `ch01d-physical-metrics.tex`, private local source. Labels `v21c01:physical-gram-definition`, `v21c01:metric-symmetry`, `v21c01:residual-decomposition`, `v21c01:gram-equivalence`.

[4] M. Subbarao, *Ram Calculus*, Volume 21, `ch01f-spectral-limits.tex`, private local source. Labels `v21c01:pollution-proposition`, `v21c01:strong-not-norm`; subsection "Domain-aware approximate inversion."

[5] M. Subbarao, *Ram Green Operators and Sparse Tree Moments for Local Heights on Semistable Curves*, `TR-meta5-math.tex`, private local source. Equations `eq:gauge`, `eq:gaugeinv`, `eq:transport`, `eq:elimpair`; selected read scope in Appendix A.

[6] N. Berestycki and E. Gwynne, *Random walks on mated-CRT planar maps and Liouville Brownian motion*, Communications in Mathematical Physics 395 (2022), 773–857. arXiv:2003.10320; DOI.

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